

Total No. of Questions : 5]

SEAT No. :

P3348

[Total No. of Pages : 4

[6027]-31

Second Year M.C.A. (Management)
IT-31 : MOBILE APPLICATION DEVELOPMENT
(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Figures to the right indicates full marks.

Q1) Multiple choice Questions. For each of the multiple type questions, choose the correct answer. **[10]**

- a) Which section provides a key component called Dalvik virtual machine.
 - i) Libraries
 - ii) Android Runtime
 - iii) Application Framework
 - iv) Application
- b) Android is based on which kernel?
 - i) Linux
 - ii) Windows
 - iii) Mac
 - iv) Redhat
- c) Choose the correct option regarding activity in android
 - i) An activity is a window that contains the user interface of your application
 - ii) An Application can have zero or more activities
 - iii) An application can have only one activity
 - iv) Option i & ii are correct
- d) Which of the following is a dialog class in android
 - i) AlertDialog
 - ii) DatePicker Dialog
 - iii) Progress Dialog
 - iv) All the above
- e) What is manifest xml in android
 - i) It has information about layout in an application
 - ii) It has all information about an application
 - iii) It has information about activities in an application
 - iv) It has information about Java classes in an application
- f) What does ES6 stands for ?
 - i) ECMAScript 6
 - ii) ECMA 6
 - iii) ECMA JavaScript 6
 - iv) E Javascript 6

P.T.O.

- g) Which of the following is built in database of android
- i) SQLite
 - ii) MySQL
 - iii) Oracle
 - iv) Mango Db
- h) Which of the following class in android displays information for a short period of time and disappear after sometime
- i) Toast class
 - ii) Log class
 - iii) Maketest class
 - iv) Main class
- i) Which method is called when SQLite database needs to be upgraded
- i) on update ()
 - ii) onupgrade ()
 - iii) onupgrade Data ()
 - iv) UpdateAll ()
- j) _____ means it will occupy the complete space available on the display of the device
- i) wrap-content
 - ii) match-content
 - iii) match-parent
 - iv) wrap-parent
- k) What method you should override to use Android menu system?
- i) On create option Menu ()
 - ii) On create Menu ()
 - iii) On create context Menu ()
 - iv) On Menu created ()
- l) What is Dart?
- i) Both ii and iii
 - ii) Dart is a object-oriented programming Language
 - iii) Dart is used to create a frontend user interfaces
 - iv) None of the above
- m) Which of the following command is used to create a react app?
- i) Install -g create-react-app
 - ii) npm install create-react-app
 - iii) npx create-react-app veactapp
 - iv) None of the above
- n) The method that is called to clean up any Resources used by service is _____
- i) Stopself ()
 - ii) Onstop ()
 - iii) On Destroy ()
 - iv) Stop Service ()

- o) RPC in web services stands for
 - i) remote procedure calls
 - ii) representation procedure call
 - iii) reactive produce call
 - iv) reductant procedure call
- p) The extras can be set and read using the?
 - i) Sets Extras () and get Extras () methods
 - ii) Puts Extras () and gets Extras () methods
 - iii) Sets Extras () and gets Extras () methods
 - iv) Put Extras () and get Extras () methods
- q) In which directory XML Layout Files are stored.
 - i) /assets
 - ii) /Src
 - iii) /res/values
 - iv) /res/layout
- r) To display text which control will be used.
 - i) EditText
 - ii) Textview
 - iii) Label
 - iv) View Text
- s) A _____ is a piece of an activity which enables more modular activity design
 - i) Fragment
 - ii) Intent
 - iii) Filters
 - iv) Sub-Activity
- t) AAPT Stands for
 - i) Android Activity packaging Tool
 - ii) Android Asset Packaging Tool
 - iii) Android Action Packaging Tool
 - iv) None of the above

Q2) Explain window Architecture with a neat diagram [6]

OR

Explain Android Architecture with a neat diagram [6]

Q3) a) Write an android application to display Alert Dialog Box on click of a button [6]

b) What is Layout? Explain any one layout with suitable example [6]

OR

a) What is Intent? Explain types of intent with a suitable example [6]

b) Demonstrate Array Adapter using List view to display list of fruitsname [6]

Q4) a) Write an android using SQLite to create a table student (Sid, sname, scourse, smark) and update a record of student [6]

b) What is Android JSON parsing explain with suitable example [6]

OR

a) Write short note on Content value with example [6]

b) Write android code to parse the data from JSON Array and display it in a Toast [6]

Q5) a) Write steps to build simple Flutter application [6]

b) Write a short note on Android Project Folder structure [4]

OR

a) What is React Native? State the advantages of React Native [6]

b) What is view? Explain any two view with suitable example [4]



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SEAT No. :

P3349

[Total No. of Pages : 5

[6027]-32

Second Year M.C.A. (Management)

IT-32 : DATA WAREHOUSING AND DATA MINING

(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Draw neat labelled diagrams wherever necessary.*

Q1) Answer the following multiple-choice questions.

[20×0.5]

- i) Generation of concept hierarchy who's depends on?
 - a) Static or dynamic data set
 - b) Only several relations
 - c) System call mechanism
 - d) Basic terminology
- ii) Which cache is use in connected and unconnected transformations?
 - a) Dynamic cache and static cache
 - b) Content Delivery Network cache
 - c) Database cache
 - d) Import cache & File system cache
- iii) Which is the Function used for K-means clustering?
 - a) K-mean
 - b) Heatmap
 - c) K-means
 - d) K-medians
- iv) What to use in Hierarchical clustering for finding the right number of clusters?
 - a) The Elbow method
 - b) Decision Trees
 - c) Dendrograms
 - d) Histograms
- v) _____collects all the information and the subjects about an entire organization.
 - a) Data mart
 - b) Virtual warehouse
 - c) Data warehouse view
 - d) Enterprise warehouse

P.T.O.

- vi) Which is not a valid layer in Three-layer data warehouse architecture in conceptual view?
- a) Processed data layer b) Real-time data layer
 - c) Derived data layer d) Reconciled data
- vii) Which is true about star schema?
- a) Suited for operational data processing
 - b) Used in operational systems
 - c) Used in operational data stores
 - d) Used to develop data warehouses & dimensional data marts
- viii) Dimensionality refers to
- a) Cardinality of key values in a star schema
 - b) The data that describes the transactions in the fact table
 - c) The level of detail of data that is held in the fact table
 - d) The level of detail of data that is held in the dimension table
- ix) Data transformation includes _____
- a) A process to change data from a detailed level to a summary level
 - b) A process to change data from a summary level to detailed level
 - c) Joining data from one source into various sources of data
 - d) Separating data from one source into various sources of data
- x) A cross tab section with one attribute is obtaining by using
- a) Slice b) Dice
 - c) Pivot d) Both slice and dice
- xi) What is true about data mining?
- a) Data mining is defined as the procedure of extracting information from huge sets of data
 - b) Data mining also involves other processes such as data cleaning, data integration, data transformation
 - c) Data mining is the procedure of mining knowledge from data
 - d) All of the mentioned

- xii) In a hypercube, each dimension belongs to _____ only.
- a) One cube
 - b) Multi cube
 - c) None of the mentioned
 - d) Both a and b
- xiii) Parametric Data Reduction involves
- a) Regression
 - b) Classification
 - c) Clustering
 - d) Association
- xiv) _____ is the application of data mining techniques to discover patterns from the web.
- a) Text mining
 - b) Multimedia mining
 - c) Web mining
 - d) Link mining
- xv) Google PR checker is a tool designed for _____
- a) Page Rank
 - b) Hits
 - c) Plagiarism
 - d) Summarization
- xvi) For KNN, which of the following distance measure is NOT valid for continuous variables?
- a) Manhattan distance
 - b) Hamming distance
 - c) Euclidean distance
 - d) Minkowski distance
- xvii) A collection of databases that offer a unified approach for organizing data and classifying data according to subject.
- a) Data mart
 - b) Enterprise data warehouse
 - c) Operational data store
 - d) Data mining
- xviii) _____ is a base document for data extraction.
- a) Logical data map
 - b) FP Tree
 - c) Granularity
 - d) Association
- xix) Microsoft SQL server is an example of
- a) HOLAP
 - b) MOLAP
 - c) ROLAP
 - d) SOLAP
- xx) Propagate the updates from the data sources to the warehouse.
- a) Data Load
 - b) Refresh
 - c) Data Transformation
 - d) Data cleaning

Q2) a) Define Data warehouse & explain architecture of data warehouse with neat diagram. [5]

b) Explain different data warehouse schemas. [5]

OR

Q2) a) Compare OLAP and OLTP. [5]

b) Explain kimball life cycle diagram in detail. [5]

Q3) a) Explain different steps involved in cleaning & transformation of data. [5]

b) What is text-mining? Explain applications of text-mining. [5]

OR

Q3) a) Brief OLAP operations with proper example. [5]

b) Explain data mining architecture. [5]

Q4) a) Apply FP-Growth tree algorithm to construct FP-Tree & find the frequent item sets. Consider following data set with minimum support 30%. [5]

TID	List of Items
1	Apple, Mango, Cocount, Banana
2	Banana, Berries, Coconut, Mango, Chikoo
3	Berries, Coconut, Dates, Grapes
4	Watermelon, Coconut, Mango
5	Apple, Banana, Chikoo, Dates
6	Apple, Berries
7	Apple, Coconut, Grapes, Banana
8	Mango, Watermelon, Coconut
9	Banana, Mango, Coconut

b) Consider the data set from **Q4) a)** and calculate support & confidence of following item sets. [5]

i) { Apple, Coconut }

ii) { Apple, Coconut, Banana }

iii) { Mango, Watermelon }

iv) { Coconut, Berries }

v) {Banana, Mango, Coconut }

OR

- Q4) a)** Consider following data set and find the frequent item sets with minimum support count 2 using Apriori Algorithm. [5]

TID	Items
T1	Hot Dogs, Buns, Ketchup
T2	Hot Dogs, Buns
T3	Hot Dogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	Hot Dogs, Coke, Chips

- b) Consider same data set in **Q4) a)** & generate association rules [minimum 3] [5]

- Q5) a)** Write K-mean clustering algorithm and apply on the following data set to group it into two clusters. [5]

$D = (0.5, 0.7, 0.9, 0.6, 0.4, 0.3, 0.10, 0.12, 0.14, 0.16, 0.35, 0.36, 0.11, 0.45, 0.58)$

- b) How KNN algorithm work? [5]

OR

- Q5) a)** $D = \{ 14, 13, 15, 16, 16, 19, 20, 35, 37, 39, 40, 41, 50, 56, 72, 17, 18, 12, 24, 22, 21 \}$

$K = 3$ $C_1 = 13$, $C_2 = 20$, $C_3 = 50$

Apply K-mean clustering. [5]

- b) Explain Agglomerative and divisive clustering mechanism. [5]



Total No. of Questions : 5]

SEAT No. :

P3350

[Total No. of Pages : 2

[6027]-33

M.C.A. - II (Management)

IT - 33 : Software Testing and Quality Assurance

(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*

Q1) a) Write a detailed test plan for online grossary ordering mobile application. The application functionalities are registration, login, check items, price, delivery time etc. **[6]**

b) Write the test cases for above application. **[4]**

Q2) a) Write the test cases for testing username and password to use email account. **[5]**

b) Write data flow control analysis for following. **[5]**

i) var i, x, sum

ii) i = 0

iii) sum = 0

iv) Input (x)

v) while (x != 0)

vi) sum = sum + x

vii) i = i + 1

viii) Input x

ix) End while

x) print (sum)

Q3) What do you mean by software quality assurance? Explain SQA activities.[10]

OR

Explain s/w quality factors and software quality metrics.

[10]

P.T.O.

- Q4)** a) Explain verification and validation in software testing. [5]
b) What are levels of testing? Explain in detail. [5]

OR

- a) What is Test Driven Development (TDD)? How TDD is performed? [5]
b) Compare Black Box and White Box testing. [5]

Q5) Write short notes on (any two): [10]

- a) Project Risk
- b) Product Risk
- c) Selenium
- d) Appium
- e) Incident Management



Total No. of Questions : 5]

SEAT No. :

P3351

[6027]-34

[Total No. of Pages : 5

Second Year M.C.A. (Management Faculty)
IT - 34 : KNOWLEDGE REPRESENTATION
AND ARTIFICIAL INTELLIGENCE : ML, DL
(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *For MCQ select appropriate choice from options given.*
- 3) *From Q.2 to Q.5 having Internal choice.*
- 4) *Figures to the right indicate full marks.*

Q1) Multiple Choice Questions:

[20×½=10]

- a) Which of the following can improve the performance of AI Agent?
 - i) Perceiving
 - ii) Learning
 - iii) Observing
 - iv) Thinking Irrationally
- b) Which of the following is not a limitation of AI?
 - i) High Cost
 - ii) Lacking Creativity
 - iii) High Accuracy
 - iv) Unemployment
- c) In Wumpus world the knowledge based agent start from location
 - i) Room [1, 1]
 - ii) Room [2, 2]
 - iii) Room [1,2]
 - iv) Room [4,4]
- d) The symbolization for a conjunction is:
 - i) $p \rightarrow q$
 - ii) $p \wedge q$
 - iii) $p \vee q$
 - iv) $\sim p$
- e) In propositional logic, given P and $P \rightarrow Q$, we can infer
 - i) $\sim Q$
 - ii) $P \wedge Q$
 - iii) Q
 - iv) $\sim P \wedge Q$
- f) Resolution refutation method removes which logical symbol from given expression.
 - i) $\wedge$ (AND)
 - ii) $\vee$ (OR)
 - iii) $\Rightarrow$ Implication
 - iv) $\neg$ (NOT)

P.T.O.

- g) Which distance metric (s) are suitable for categorical variables to find the closest neighbors.
- i) Euclidean Distance
 - ii) Manhattan Distance
 - iii) Minkowski Distance
 - iv) Hamming Distance
- h) Which of the following measures is optional for comparing the goodness of the fit of competing regression models involving the same dependent variable?
- i) The Intercept
 - ii) The coefficient
 - iii) R-square
 - iv) Standard Deviation of the residuals
- i) Logistic regression is a _____ regression technique that is used to model data having a _____ outcome.
- i) Linear, Numeric
 - ii) Linear, Binary
 - iii) Nonlinear, Numeric
 - iv) Nonlinear, binary
- j) The correlation coefficient for two real-valued attributes is 0.85. What does this value tell?
- i) The attributes are not linearly related
 - ii) As the value of one attribute increases the value of second attribute also increases
 - iii) As the value of one attribute decrease the value of second attribute increases
 - iv) The attribute show a curvilinear relationship
- k) In model based learning methods, an iterative process take places on the ML models that are built, based on various model parameters called?
- i) Mini-Batches
 - ii) Optimized parameters
 - iii) Hyper parameters
 - iv) Super parameters
- l) In which of the following learning the teacher returns reward and punishment to learner?
- i) Active learning
 - ii) Reinforcement learning
 - iii) Supervised learning
 - iv) Unsupervised learning

- m) The most common Neural networks consist of_____ network layers.
- i) 1
 - ii) 2
 - iii) 3
 - iv) 4
- n) Which layer is the most important layers where features extraction take place, and adjustments are made to train faster and function better.
- i) Input layer
 - ii) Output layer
 - iii) Hidden layer
 - iv) None of these
- o) Which strategy does not prevent a model from over-fitting to the training data?
- i) Early stopping
 - ii) Dropout
 - iii) Data Augmentations
 - iv) Pooling
- p) Weight sharing occurs in which neural network architecture?
- i) CNN
 - ii) RNN
 - iii) Fully connected Neural Network
 - iv) CNN & RNN
- q) ANN used for:
- i) Pattern Recognition
 - ii) Classification
 - iii) Clustering
 - iv) All of these
- r) It is the difference between the validation error and training error.
- i) Std Dev
 - ii) Variance
 - iii) Dispersion
 - iv) Bias
- s) Chatbot is based on which AI technique?
- i) Big data
 - ii) Robotics
 - iii) Deep learning
 - iv) NLP
- t) What are the devices that sense the physical environments?
- i) Control unit
 - ii) Sensors
 - iii) CPU
 - iv) Firmware

Q2) a) What is knowledge representation? Explain the different techniques of KR. [4]

b) Consider the following. [6]

“As per the law, it is a crime for an American to sell weapons to hostile nation country an enemy of America, has some missiles, & all the missiles were sold to it by Robert, who is an American Citizen” Prove that “Robert is Criminal” using forward chaining algorithm.

OR

a) Why do we need Artificial Intelligence? [4]

b) Write a FOL of following statement. [6]

i) Mary loves everyone.

ii) No one talks.

iii) Everyone loves everyone.

iv) Everyone loves everyone except himself.

v) Someone loves everyone.

vi) Someone walks and someone talks.

Q3) a) State the Mathematical formulation of the SVM. [4]

b) Suppose 10000 patients get tested for covid-19, out of them, 9000 are actually healthy and 1000 are actually sick. For the sick people, a test was positive for 620 and negative for 380. For healthy people, the same test was positive for 180 and negative for 8820. Construct a confusion matrix for data and compute precision & recall for the data. [6]

OR

a) How SVM can be used for classification of linearly separable data? [4]

b) Use K-means clustering to cluster the following data into two groups. Assume cluster centroid are $m_1 = 2$ and $m_2 = 4$. The distance function used is Euclidean distance.

{2, 4, 10, 12, 3, 20, 30, 11, 25} [6]

Q4) a) Explain the use of Long Short Term Memory (LSTM). [4]

b) Why do we use pooling layers in CNN? [4]

c) Explain ANN. [2]

OR

a) Explain Convolutional Neural Network (CNN). [4]

b) Why do we prefer CNN over ANN for image data as input? [6]

Q5) Write a short notes on

[10]

- a) Issues in knowledge representation.
- b) Random forest.
- c) Building Block of DL.
- d) FPGA
- e) Speech recognition.

OR

- a) List out type of AI.
- b) Advantage of Logistic Regression.
- c) Detection in Deep learning.
- d) GPU
- e) Chatbot.



Total No. of Questions : 5]

SEAT No. :

P3352

[Total No. of Pages : 4

[6027]-35

S.Y. M.C.A. (Management)
IT - 35 : CLOUD COMPUTING
(2020 Pattern) (Semester -III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Figures to the right indicate full marks.*

Q1) Write correct option (or) filling the blanks of following questions.*[each ½ Mark]*

- a) Broadly divide the cloud Architecture into?
 - i) 2
 - ii) 3
 - iii) 4
 - iv) 5
- b) Which of the following is Type - 1 Hypervisor?
 - i) Windows River simics
 - ii) Virtual server 2005 R2
 - iii) KVM
 - iv) Lynx secure
- c) Which of the following runs on Xen Hypervisor?
 - i) Azure
 - ii) AWS E2
 - iii) AWS EC3
 - iv) All of the mentioned
- d) Which of these is Identity in I AM?
 - i) Groups
 - ii) Users
 - iii) Roles
 - iv) All of the mentioned
- e) Which of the following services that need to be negotiated in service Level Agreement ?
 - i) Logging
 - ii) Auditing
 - iii) Regulatory compliance
 - iv) All of the mentioned
- f) The technology used to distribute service requests to resources is referred to as _____.
 - i) CND
 - ii) CDN
 - iii) CWD
 - iv) All of the mentioned

P.T.O.

- g) IAM group :
- i) Is same as IAM users
 - ii) Can be used to specify permission for a collection of users
 - iii) Is truly an identity
 - iv) All of these
- h) Which of the following is more sophisticated load balancer ?
- i) Workload managers
 - ii) Workspace managers
 - iii) Rackspace managers
 - iv) None of these
- i) Which of the following service provider provides the least amount of built in security.
- i) SAAS
 - ii) PAAS
 - iii) IAAS
 - iv) All of the mentioned
- j) Which of the following combination is possible in websphere ESB?
- i) XML/JMS
 - ii) SOAP/JMS
 - iii) SOAP/HTTP
 - iv) Governance
- k) What is the correct formula to calculate the cost of a cloud computing development
- i) $\text{Cost CLOUD} = \sum (\text{unit cost CLOUD} / \text{Revenue} + \text{cost CLOUD})$
 - ii) $\text{Cost CLOUD} = \sum (\text{unit cost CLOUD} / \text{Revenue} - \text{cost CLOUD})$
 - iii) $\text{Cost CLOUD} = \sum (\text{unit cost CLOUD} \times \text{Revenue} - \text{cost CLOUD})$
 - iv) None of the mentioned.
- l) What are the features of docker
- i) Easy Modeling
 - ii) Version control
 - iii) Placement /Affinity
 - iv) All of the mentioned
- m) Docker was developed in ____ .
- i) 2012
 - ii) 2014
 - iii) 2013
 - iv) 2015
- n) How many different protocol types are required for message passing in SOA.
- i) 2
 - ii) 4
 - iii) 1
 - iv) 3

- o) What ports does Docker registry use ?
- i) Port 3000
 - ii) Port 5000
 - iii) Port 8000
 - iv) None of the mentioned
- p) An application that provides for transaction overflow in a reservation system is an example of ____.
- i) Cloud computing / bursting
 - ii) Cloud provisioning
 - iii) Cloud servicing
 - iv) All of the mentioned
- q) Which of the following software can be used to implement load balancing?
- i) Apache mod - balancer
 - ii) Apache mod-proxy- balancer
 - iii) F6's BigIP
 - iv) All of the mentioned
- r) Which of the following essential object in WSDL is used to support message transfer?
- i) Bus
 - ii) Type
 - iii) Binding
 - iv) None of the mentioned
- s) Data stored in ____ domains doesn't require maintenance of a schema.
- i) "Simple DB"
 - ii) SQL server
 - iii) Storage
 - iv) Hardware
- t) Implementing ____ scanner is a tool that can scan an entire system for known vulnerabilities.
- i) Malware
 - ii) Vulnerability
 - iii) Worm
 - iv) Virus

Q2) Describe in detail about cloud service Models & Deployment models with example's. **[10]**

OR

Describe in detail about cloud computing, cluster computing & Grid computing with suitable example. **[10]**

Q3) a) What is Hypervisor & explain its types with example. [5]

b) Write a short note on service oriented Architecture [5]

OR

a) Differentiate between SOAP and Restful webservices (REST) [5]

b) Write a short note on service level Agreement. [5]

Q4) a) Differentiate between Multi - core Architecture & Multi - cloud Environment. [5]

b) Write a short note on cloud computing Architecture . [5]

OR

a) What is parallel programming? [5]

b) Explain cloud Migration strategies & process. [5]

Q5) Suggest the cloud computing Architecture to be implemented by pharma Industry. [10]

OR

How to maintain a Disaster situation in cloud & Explain the Concept in details. [10]

